

A2 HW

Thursday, September 10, 2026 12:00 PM

Q1

P1.1 Show that $m(a+bX) = a + b \cdot m(X)$

$$m(a+bX) = \frac{1}{N} \sum_{i=1}^N (a + bx_i) = \frac{1}{N} \left(\sum_{i=1}^N a + b \sum_{i=1}^N x_i \right) = \frac{1}{N} (Na) + b$$

$$= \boxed{a + b \cdot m(X)}$$

P1.2

show that $\text{cov}(X, a+bY) = b \cdot \text{cov}(X, Y)$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) [(a + by_i) - m(a + bY)]$$

$$\hookrightarrow (a + by_i) - (a + b \cdot m(Y)) = b(y_i - m(Y))$$

$$\hookrightarrow \text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) \cdot b(y_i - m(Y)) = b \cdot \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y))$$

$$= \boxed{b \cdot \text{cov}(X, Y)}$$

P1.3

show that $\text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$ and in particular that $\text{cov}(X, X) = s^2$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N [(a + bx_i) - m(a + bX)] [(a + bx_i) - m(a + bX)]$$

$$\hookrightarrow \text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N b(x_i - m(X)) \cdot b(x_i - m(X)) = b^2 \cdot \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

$$\hookrightarrow \text{cov}(X, X) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(x_i - m(X)) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

$$\hookrightarrow \text{cov}(a+bX, a+bX) = b^2 \cdot \text{cov}(X, X)$$

$$\hookrightarrow \boxed{s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2 = \text{cov}(X, X)}$$

P1.4

consider a non-decreasing transformation $g(\cdot)$.
Is it always true that $m(g(X)) = g(m(X))$?

$\hookrightarrow$ No, this is only true in general when g is a linear function
for example when $g(x) = a + bx$, but for a general
non-decreasing but nonlinear g , the equality would fail and
wouldn't be true anymore.

Another example is, $g(x) = x^2$ with the data 2, 4.

Squaring first then averaging results in 10

$$2^2 + 4^2$$

$$\sqrt{4} + \sqrt{16} = \frac{20}{2} = 10$$

while averaging first then squaring results in 9.

$$\sqrt{2 + 4}$$

$$\frac{6}{2} = 3^2 \rightarrow 9$$

The values don't match. Nonlinear functions like squaring stretch the large values more than small ones so the order of averaging vs. transforming matters.